SHUFAN SHAHI

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Education

Islamic University of Technology **2022–2026**
B.Sc. in Computer Science and Engineering **CGPA: 3.58**
Coursework: Algorithms, Machine Learning, Deep Learning, Databases, and Networks.

Chuadanga Govt. College **2021**
Higher Secondary Certificate (HSC) **GPA: 5.00/5.00**

Experience

AI Engineer **2026–Present**
GLiTch LABS

- Built RAG architectures, connecting LLMs to vector databases for optimized semantic search.

Student Researcher **2026**
Elite Research Lab

- Support research projects through literature reviews, experiments, and analysis.

Intern **2025**
Battery Low Interactive Ltd.

- Learned Flutter development, REST API integration, and applications of fine-tuned ML models.

Researcher **2025–2026**
IUT Computer Vision Lab

- Researched multimodal AI methods to improve integration and mitigate modality imbalance.

Senior Executive **2025–2026**
Project Aero

- Contributed to drone R&D by applying AI-based object tracking for international UAV competitions.

Projects

DocuMind - Document Q&A | *Python, LangChain, LlamaIndex, ChromaDB, Ollama* GitHub

- Built a local RAG system for documents with cited answers and hallucination guardrails.
- Compared 4 chunking methods and 2 embedding models; added hybrid BM25 and reranking.

Benhallu - Bengali Hallucination Detection | *Python, RAG, BanglaBERT, scikit-learn* GitHub

- Built a retrieval-augmented pipeline to verify Bengali answers against supplied context or a Bengali QA knowledge base.
- Created a hallucination dataset and used fine-tuned BanglaBERT with answer-comparison features to classify responses.

Bengali Speaker Diarization - Speaker Turn Detection | *PyTorch, Silero VAD, ECAPA-TDNN* GitHub

- Built a pipeline using neural voice activity detection, speaker embeddings, and agglomerative clustering to label timestamped speaker turns.
- Estimated speaker counts with silhouette scores and exported diarization results to CSV.

Drone Vision - Aerial Tracking | *PyTorch, YOLOv11X, ByteTrack* GitHub
- Fine-tuned YOLOv11X on VisDrone and used ByteTrack for human counting and multi-object tracking.

Biztrack - Business Analytics | *Next.js, Node.js, PostgreSQL* GitHub Demo
- Built AI modules for real-time finance tracking, risk alerts, customer segmentation, and forecasting.

Postfolio - Professional Networking | *Spring Boot, Next.js, Docker, RabbitMQ, Stripe* GitHub Demo
- Integrates social networking, portfolio and CV creation, job search, and AI interview preparation.

Redrick Routinson - Campus Automation | *Python, React, Flask, Firebase* GitHub Demo
- Built GA scheduling, QR exam attendance, class swapping, events, forums, and lost-and-found.

Research Experiences

EthMuSAM: Mutual Scoring with SAM for Zero-Shot Anomaly Segmentation **2026**
- Combined DINOv2 patch embeddings with SAM-based mutual scoring to localize zero-shot anomalies.

Prototype-Guided Adaptive Feature Learning for Multimodal Emotion Recognition in Conversation (MERC) **2026**
- Used adaptive weighting and modality prototypes to improve multimodal emotion recognition.

Rethinking Visual Features in Graph-Based Multimodal Emotion Recognition in Conversations (MERC): From Low-Level Pixels to Semantic Representations **2026**
- Used semantic visual representations to improve graph-based multimodal emotion recognition.

A Cross-Lingual Information Retrieval System for Bangla–English News Corpora **2025**
- Developed a hybrid CLIR system using lexical, fuzzy, and embedding-based retrieval.
- Handled cross-language and code-switched queries with semantic ranking.

Weakly Supervised Named Entity Recognition via Sparse-CHMM and BERT **2025**
- Aggregated noisy labeling functions without ground-truth annotations.
- Modeled sequence dependencies using Sparse-CHMM with contextual embeddings.

Technical Skills

Programming Languages: Python, C++, Java
Data and Machine Learning: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas
Databases: MySQL, Firebase, PostgreSQL
Web Development: React, Tailwind CSS, Next.js, Spring Boot
Mobile Development: React Native, Flutter
Visualization: Matplotlib, Seaborn
Development Tools: Git, Linux

Achievements, Awards and Scholarships

- **5th Place, VAND4 International Competition** — Zero-Shot Industrial Track **2026**
- **First Runner-up, IUT Coderush Hackathon** — Rapid team application development **2025**
- **7th Place, BAIC Datathon** — Data preprocessing, visualization, and modeling **2025**

Reference

Shahriar Ivan
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